Publications

The asterisks (*) mark the corresponding authors.

31 D. Picconi*, M. S. F. Menger, E. Palacino-González, E. X. Salazar and S. Faraji*, **Implementation of quasiclassical mapping approaches for nonadiabatic molecular dynamics in the PySurf package**, *Phys. Chem. Chem. Phys.*, Accepted Manuscript (2025)

30 L. E. Cigrang, B. F. E. Curchod, R. A. Ingle, A. Kelly, J. R. Mannouch, D. Accomasso, A. Alijah, M. Barbatti, W. Chebbi, N. Došlić, E. C. Eklund, S. Fernandez-Alberti, A. Freibert, L. González, G. Granucci, F. J. Hernández, J. Hernández-Rodríguez, A. Jain, J. Janoš, I. Kassal, A. Kirrander, Z. Lan, H. R. Larsson, D. Lauvergnat, B. Le Dé, Y. Lee, N. T. Maitra, S. K. Min, D. Peláez, D. Picconi, U. Raucci, Z. Qiu, P. Robertson, E. Sangiogo Gil, M. Sapunar, P. Schürger, P. Sinnott, S. Tretiak, A. Tikku, P. Vindel-Zandbergen, G. A. Worth, F. Agostini*, S. Gómez*, L. M. Ibele* and A. Prlj*, **Roadmap for molecular benchmarks in nonadiabatic dynamics**, *J. Phys. Chem. A* (2025)

29 E. Palacino-González* and D. Picconi*, **Photoinduced Molecular Quantum Dynamics: Theory and computational Methods**, in *Quantum dynamics and spectroscopy of functional molecular materials and biological photosystems*, J. Léonard & T. Renger (eds.), *EDP Sciences*, in press (2025)

28 G. Pacella, M. Nabatova, Y. Zhang, D. Picconi, R. R. Weber, S. Faraji and G. Portale*, **Harnessing negative photochromism in styryl cyanines for light-modulated proton transport**, *Angew. Chem. Int. Ed.* e202506532 (2025)

27 F. Lever*, D. Picconi, D. Mayer, S. Ališauskas, F. Calegari, S. Düsterer, R. Feifel, M. Kuhlmann, T. Mazza, J. Metje, M. Robinson, R. Squibb, A. Trabattoni, M. Ware, P. Saalfrank, T. J. A. Wolf and M. Gühr*, **Direct observation of $\pi\pi^*$ to $n\pi^*$ transition in 2-thiouracil via time resolved NEXAFS spectroscopy**, *J. Phys. Chem. Lett.* **16**, 4038 (2025)

26 D. Picconi*, **Dynamics of high-dimensional quantum systems coupled to a harmonic bath. General theory and implementation via multiconfigurational wave packets and truncated hierarchical equations for the mean-fields**, *J. Chem. Phys.* **161**, 164108 (2024)

25 D. Mayer*, M. Handrich, D. Picconi, F. Lever, L. Mehner, M. L. Murillo Sanchez, C. Walz, E. Titov, J. D. Bozek, P. Saalfrank and M. Gühr, **X-ray photoelectron and NEXAFS spectroscopy of thionated uracils in the gas phase**, *J. Chem. Phys.* **161**, 134301 (2024)

24 S. Faraji*, D. Picconi* and E. Palacino-González*, **Advanced quantum and semiclassical methods for simulating photo-induced molecular dynamics and spectroscopy**, *WIREs Comput. Mol. Sci.* **14**, e1731 (2024)

23 M. Horz, H. M. A. Masood, H. Brunst, J. Cerezo, D. Picconi, H. Vormann, M. S. Niraghatam, L. J. G. W. van Wilderen, J. Bredenbeck*, F. Santoro* and I. Burghardt*, **Vibrationally resolved two-photon electronic spectra including vibrational pre-excitation: Theory and application to VIPER spectroscopy with two-photon excitation**, *J. Chem. Phys.* **158**, 064201 (2023)

22 D. Picconi*, **Quantum dynamics of the photoinduced charge separation in a symmetric donor–acceptor–donor triad: The role of vibronic couplings, symmetry and temperature**, *J. Chem. Phys.* **156**, 184105 (2022)

21 D. Mayer, D. Picconi, M. S. Robinson and M. Gühr*, **Experimental and theoretical gas-phase absorption spectra of thionated uracils**, *Chem. Phys.* **558**, 111500 (2022)

20 D. Mayer, F. Lever, D. Picconi*, J. Metje, S. Alisauskas, F. Calegari, S. Düsterer, C. Ehlert, R. Feifel, M. Niebuhr, B. Manschwetus, M. Kuhlmann, T. Mazza, M. S. Robinson, R. J. Squibb, A. Trabattoni, M. Wallner, P. Saalfrank, T. J. A. Wolf and M. Gühr*, **Following excited-state chemical shifts in molecular ultrafast x-ray photoelectron spectroscopy**, *Nat. Commun.* **13**, 198 (2022)

19 D. Picconi*, **Nonadiabatic quantum dynamics of the coherent excited state intramolecular proton transfer of 10-hydroxybenzo[h]quinoline**, *Photochem. Photobiol. Sci.* **20**, 1455 (2021)

18 F. Lever, D. Mayer, D. Picconi*, J. Metje, S. Alisauskas, F. Calegari, S. Düsterer, C. Ehlert, R. Feifel, M. Niebuhr, B. Manschwetus, M. Kuhlmann, T. Mazza, M. S. Robinson, R. J. Squibb, A. Trabattoni, M. Wallner, P. Saalfrank, T. J. A. Wolf and M. Gühr*, **Ultrafast dynamics of 2-thiouracil investigated by time-resolved Auger spectroscopy**, *J. Phys. B: At. Mol. Opt. Phys.* **54**, 014002 (2020)

17 D. Picconi* and I. Burghardt, **Time-resolved spectra of I_2 in a krypton crystal by G-MCTDH simulations: Nonadiabatic dynamics, dissipation and environment driven decoherence**, *Faraday Discuss.* **221**, 30 (2020)

16 D. Picconi* and I. Burghardt*, **Creation and Detection of Molecular Schrödinger Cat States: Iodine in Cryogenic Krypton Observed via Four-Wave-Mixing Optics**, in *Advances in Open Systems and Fundamental Tests of Quantum Mechanics*, B. Vacchini et al. (eds.), *Springer Proceedings in Physics*

15 D. Picconi* and I. Burghardt*, **Open system dynamics using Gaussian-based multiconfigurational time-dependent Hartree wavefunctions: Application to environment modulated tunneling**, *J. Chem. Phys.* **150** 224106 (2019)

14 D. Picconi*, J. A. Cina and I. Burghardt, **Quantum dynamics and spectroscopy of dihalogens in solid matrices. II. Theoretical aspects and G-MCTDH simulations of time-resolved coherent Raman spectra of Schrödinger cat states of the embedded I_2Kr_{18} cluster**, *J. Chem. Phys.* **150**, 064112 (2019)

13 D. Picconi*, J. A. Cina and I. Burghardt, **Quantum dynamics and spectroscopy of dihalogens in solid matrices. I. Efficient simulation of the photodynamics of the embedded I_2Kr_{18} cluster using the G-MCTDH method**, *J. Chem. Phys.* **150**, 064111 (2019)

12 S. Yu. Grebenshchikov* and D. Picconi, **Entanglement of the molecular photodissociation products at avoided crossings and conical intersections**, *Chem. Phys.* **515**, 60 (2018)

11 D. Picconi* and S. Yu. Grebenshchikov*, **Photodissociation dynamics in the first absorption band of pyrrole. II. Photofragment distributions for the $^1A_2(\pi\sigma^*) \leftarrow \tilde{X}^1A_1(\pi\pi)$ transition**, *J. Chem. Phys.* **148**, 104104 (2018)

10 D. Picconi* and S. Yu. Grebenshchikov*, **Photodissociation dynamics in the first absorption band of pyrrole. I. Molecular Hamiltonian and the Herzberg-Teller absorption spectrum for the $^1A_2(\pi\sigma^*) \leftarrow \tilde{X}^1A_1(\pi\pi)$ transition**, *J. Chem. Phys.* **148**, 104103 (2018)

9 S. Yu. Grebenshchikov* and D. Picconi, **Fano resonances in the photoinduced H-atom elimination dynamics in the $\pi\sigma^*$ states of pyrrole**, *Phys. Chem. Chem. Phys.* **19** 14902 (2017)

8 J. Ehrmaier, D. Picconi, T. N. V. Karsili and W. Domcke*, **Photodissociation dynamics of the pyridinyl radical: Time-dependent quantum wave-packet calculations**, *J. Chem. Phys.* **146** 124304 (2017)

7 D. Picconi and S. Yu. Grebenshchikov*, **Partial dissociative emission cross sections and product state distributions of the resulting photofragments**, *Chem. Phys.* **481**, 231 (2016)

6 D. Picconi and S. Yu. Grebenshchikov*, **Intermediate photofragment distributions as probes of non-adiabatic dynamics at conical intersections: application to the Hartley band of ozone**, *Phys. Chem. Chem. Phys.* **17**, 28932 (2015)

5 D. Picconi and S. Yu. Grebenshchikov*, **Signatures of a conical intersection in photofragment distributions and absorption spectra: Photodissociation in the Hartley band of ozone**, *J. Chem. Phys.* **141**, 074311 (2014)

4 D. Padula, D. Picconi, A. Lami, G. Pescitelli and F. Santoro*, **Electronic Circular Dichroism in Exciton-Coupled Dimers: Vibronic Spectra from a General All-Coordinates Quantum-Dynamical Approach**, *J. Phys. Chem. A* **117**, 3355 (2013)

3 D. Picconi, F. J. A. Ferrer, R. Improta, A. Lami and F. Santoro*, **Quantum-classical effective-modes dynamics of the $\pi\pi^* \rightarrow n\pi^*$ decay in 9H-adenine. A quadratic vibronic coupling model**, *Faraday Discuss.* **163**, 223 (2013)

2 D. Picconi, A. Lami and F. Santoro*, **Hierarchical transformation of Hamiltonians with linear and quadratic couplings for nonadiabatic quantum dynamics: Application to the $\pi\pi^*/n\pi^*$ internal conversion in thymine**, *J. Chem. Phys.* **136**, 244104 (2012)

1 D. Picconi, V. Barone, A. Lami, F. Santoro, R. Improta*, **The Interplay between $\pi\pi^*/n\pi^*$ Excited States in Gas-Phase Thymine. A Quantum Dynamical Study**, *ChemPhysChem* **12**, 1957 (2011)